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GROUP REPORT

Climate change and resilient technology adoption in developing economies - the case for bundled finance solutions in Guatemala

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Group O

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1 Introduction

Ranked seventeenth globally for extreme weather vulnerability in the 2026 Climate Risk Index Report, Guatemala is one of the most vulnerable countries to climate change in the world (Adil et al., 2026). The country generates over 80% of its GDP in zones exposed to drought, flooding, and hurricanes (Adil et al., 2026). Yet, it lacks the financial infrastructure required to protect farmers at high-risk to climate shocks. This risk is concentrated among its 1.3 million smallholder farmers (82% of all agricultural holdings), with average farmland smaller than two hectares, and in fewer than 8% of cases have access to formal credit (Adil et al., 2026; FAO, 2018).

Climate risk is particularly acute in the 'Dry Corridor', a region classified as tropical dry forest or dry tropics, covering not only parts of Guatemala but also Costa Rica, Nicaragua, Honduras, El Salvador (FAO, no date). In this area, recurrent droughts, such as those associated with El Niño events, have reduced maize and bean yields by up to 30-40% in a single season (EM-DAT, 2024). Other regions face different but equally significant risks, including frost in coffee-growing highlands and hurricane exposure along the Pacific coast (Trocaire, 2025). Climate change is intensifying and exacerbating these patterns, increasing both the frequency and severity of shocks (Cabrera et al., 2018). Despite high exposure and vulnerability, effective financial insurance mechanisms and risk management measures remain almost absent in Guatemala (World Bank DRIFCA, 2024). The Ministry of Agriculture, Livestock and Food (MAGA)'s Seguro Colectivo Paramétrico, the most significant domestic initiative, reached approximately 40,000 farmers in 2022, less than 3% of the smallholder population (MAGA, 2022), showing that formal agricultural insurance barely exists at scale. Weather data collected through only 97 stations nationwide is insufficient to calibrate reliable rainfall indices at the spatial resolution that smallholder agriculture requires, while informal risk-sharing fails under covariate shocks such as drought (INSIVUMEH; World Bank DRIFCA, 2024). At the same time, access to credit is constrained. Most smallholders lack the land titles and credit histories lenders require, while weak enforcement and correlated climate default risk lead lenders to ration credit through restrictive eligibility criteria and short maturities (Wenner, 2007; Carter, Cheng & Sarris, 2016). The result is a financial system that effectively excludes smallholder farmers from investing in necessary climate-resilient technologies. Farmers face a fundamental trade-off: adopting improved seeds or practices requires upfront investment, but a single failed harvest can make repayment impossible. As a result, many remain locked into low-risk, low-productivity systems (Farrin and Miranda, 2015).

This report thus evaluates the challenges of introducing a bundled financial solution by an NGO, combining index-based crop insurance with input credit. The objective is to enable Guatemalan smallholder farmers to invest in climate resilient technologies such as drought-tolerant maize, bean and coffee seeds, and agroforestry, whilst protecting them through index-based insurance and subsidised input credit against the weather shocks that could be catastrophic to their livelihoods during the transition. To quantify these dynamics, the quantitative model developed in this report focuses specifically on bundled insurance, credit and drought-tolerant maize and bean production in the Dry Corridor, with multi-crop extension discussed as a direction for future work. The efficacy of this solution depends upon two interrelated factors. Firstly, is achieving uptake among farmers who may face liquidity constraints, limited financial literacy and low trust in formal insurance products. Secondly, managing basis risk, which occurs when index-based payouts are not the same as farmers' actual losses and has a direct effect on trust and long-term participation.

To address these challenges, A Monte Carlo simulation[1] is employed to evaluate how different design parameters such as basis risk, poor index performance, affordability and trust affect key indicator outcomes, including farmer income, adoption rates, downside risk and default rates. Ten different design configurations for the product are tested to assess trade-offs between affordability, insurer viability and sustainability of adoption over time and to examine specifically how basis risk, pricing and trust dynamics interact to determine whether sustained uptake can be achieved. The results provide evidence for how an NGO can design a product that is both financially viable and optimised for farmer uptake, whilst also being capable of supporting long-term use of the product for greater climate resilience.

[1] GitHub repository containing the full implementation of the model, along with instructions to reproduce the results and details on the experimental setup: https://github.com/PauloBrl/Bundling-Crop-Insurance-with-Innovative-Production-Technologies/tree/main

2 Conceptual Rationale and Problem Framing

2.1 The economic logic of bundling insurance and input finance

Smallholder farmers in Guatemala face two key constraints that limit investment in improved agricultural technologies: lack of access to finance and high exposure to climate risk (Farrin and Miranda, 2015). Adopting climate-resilient inputs, such as drought-tolerant seeds, requires upfront investment that exceeds the financial capacity of most smallholders (Carter, Cheng and Sarris, 2016). Formal credit is largely unavailable, and informal lending carries high interest rates reflecting systemic risk. Even when credit is accessible, farmers may rationally avoid borrowing, as climate shocks can render income insufficient to repay loans. This results in a poverty trap, where farmers remain in low-productivity systems despite the availability of higher-return technologies. The constraint is not simply financial but behavioural: the risk of catastrophic loss makes investment unattractive even when expected returns are positive. Credit and insurance individually do not resolve this problem. Credit alone increases exposure to default risk, while insurance alone does not enable investment (Carter, Cheng and Sarris, 2016). Bundling the two addresses both constraints simultaneously by enabling access to inputs while reducing downside risk (Farrin and Miranda, 2015; Biffis et al., 2022). The efficacy of this solution depends on context. Where either credit or risk constraints are weak, the benefits of bundling diminish (Carter, Cheng and Sarris, 2016). In the Guatemalan Dry Corridor, both constraints are strongly binding, making bundling particularly relevant.

2.2 Three linked mechanisms: risk transfer, credit enhancement, and adoption feedback

2.2.1 Risk transfer

In drought-prone regions such as Guatemala's Dry Corridor, the primary value of insurance lies in changing behaviour before shocks occur, rather than compensating losses after the event has occurred (Carter, Cheng and Sarris, 2016). Historical droughts (e.g. 1998, 2009, 2015) have reduced maize yields by 30-40%, making investment in improved inputs highly risky (EM-DAT, 2024; FAOSTAT, 2024). By guaranteeing a payout in the event of a drought, insurance enables farmers to adopt higher-risk, higher-return technologies, such as drought-tolerant seeds, without maintaining low-input fallback strategies. This "ex-ante adoption effect" is well established in the literature (Carter, Cheng and Sarris, 2016). Evidence from a WFP-Aseguradora Rural pilot in Alta Verapaz and Chiquimula, which reached over 14,000 farmers with low premiums (USD 23) and meaningful payouts (up to USD 309), demonstrates that parametric insurance can be delivered at scale and influence investment decisions (WFP, 2022). This mechanism is constrained by basis risk. Index insurance pays based on a climatic indicator rather than realised farm losses, and in Guatemala, sparse meteorological infrastructure limits accurate calibration (INSIVUMEH; World Bank DRIFCA, 2024). As a result, farmers may experience substantial losses without triggering a payout. This "false negative" scenario where a farmer loses roughly 35% of output but still repays the loan, can undermine trust and reduce future uptake. Managing this trade-off between cost-efficient parametric insurance and basis risk which can damage trust is a central design challenge to this bundled product.

2.2.2 Credit enhancement

Insurance also improves access to credit by reducing lenders' exposure to covariate risk. In Guatemala, drought can trigger widespread default because drought hits every farmer simultaneously, leaving lenders with zero diversification, making agricultural lending inherently risky and leading to high interest rates often above 20% (Farrin and Miranda, 2015). By covering systemic weather risk, index insurance reduces expected losses for lenders and can lower the risk premium embedded in loan pricing. For example, reducing interest rates from 20% to 16% on a USD 350 input loan lowers annual repayment costs from USD 70 to USD 56, a meaningful saving relative to smallholder income (Farrin and Miranda, 2015; BANRURAL, 2022). Despite improved credit conditions, access remains inhibited by eligibility requirements such as collateral, credit history, or cooperative membership. In Guatemala, many smallholders operate on informal or communal land without formal titles, limiting their ability to access bundled finance (Carter, Cheng and Sarris, 2016). As a result, while insurance can make lending more viable by reducing risk for lenders, it does not automatically ensure inclusion of the most vulnerable farmers.

2.2.3 Adoption feedback

Over time, the adoption of climate-resilient technologies can reduce farmers' exposure to weather risk. As production becomes less sensitive to rainfall variability, insurance losses decline, improving the insurer's loss ratio and creating scope for lower premiums (Biffis et al., 2022). This generates a positive feedback loop: adoption reduces risk, lower risk reduces insurance costs and lower costs increase affordability and uptake. If continued, this dynamic can make the bundled product progressively more self-sustaining. However, this process also undermines the data used to price insurance. As farmers shift from traditional to improved technologies, historical climate-yield relationships become less accurate. Without regular recalibration, insurers risk mispricing contracts, underestimating risk in some conditions and overestimating it in others and thereby threatening both financial sustainability and farmer trust (Biffis et al., 2022; Biffis and Chavez, 2017).

3 Demand-Side Challenges: Smallholder Farmers

As a core stakeholder in this context, Guatemalan smallholder farmers face several types of demand-side challenges, limiting their ability to uptake bundled financing options and adopt innovative agricultural technologies. This is due to limited farmer understanding, trust and access to these finance bundles, crucially underpinned by the existing agri-finance structures and location-specific context. These barriers often interface and compound, severely limiting the agency of NGOs and governments in implementing and assisting these solutions, respectively.

3.1 Feasibility & Eligibility

Successful bundled solutions are highly dependent on their accessibility to those who require them most. In this case, it would be poor family farmers that are vulnerable to adverse climatic impacts, 70% of which live below the poverty line in Northern Central America (World Bank, 2024). These financing solutions can derisk credit channels for these farmers via their index insurance lever, however this has been demonstrated with Tanzanian maize farmers to reduce, not necessarily eliminate the need for additional informational and financial assurances, such as credit histories and collateral assets (Biffis et al., 2022). If these novel bundles are unreliable or not sufficiently trusted by lenders due to their nascent nature, this insurance may be inadequate to unlock credit. Further requirements push the risk back onto the farmers critically when their financial resources are lowest, further stripping them of their assets and inventory after failed repayment.

Beyond these regressive impacts, insufficient insurance backing could also result in social exclusion, as with requested credit history. Smallholder farmers practicing traditional methods may have previous difficulties with credit tools or lack prior access to these instruments, providing the lenders with little alternative but to deny them access to innovative financing opportunities. Furthermore, when identifying insurance payees, land titles, ownership and boundaries are also usually required as to ensure potential compensation is sufficient but not overdistributed. Traditional agricultural networks and indigenous communities that commonly practice on communally occupied land, sometimes with no formal ownership complicates this process, potentially increasing the excluded population (World Bank, 2016).

A form of indirect exclusion from these initiatives may be through poor financial literacy and understanding, especially for these smallholder farmers with little exposure to complex financial instruments. Index insurance already complicates this interaction beyond simpler and more commonly used indemnity schemes and when these are overlaid with various credit tools with financial and physical disbursement mechanisms, these bundles can be difficult to understand and trust, especially if unobservable. With under 40% of adults in Northern Central America being financially literate and rural citizens being less likely to be financially included, NGOs looking to integrate into Guatemala will face complicated stakeholder interactions, requiring educational assistance schemes (World Bank, 2024).

Finally, individual viability is essential for these farmers as they need to be financially able to repay these loans. Even if the insurance can derisk the loans, this does not directly mean that they are unlocked for the farmers themselves as interest rates that remain too high, or are perceived to be,

will hinder bundle and technological adoption rates given the excessive individual risk of failed repayment. Existing insurance mechanisms integrated in the Guatemala MAGA Seguro Colectivo Paramétrico Program are strong foundations on which to add credit channels, but these must be financially feasible for farmers.

If successful in early year implementation, trust between smallholder farmers, lenders and insurers can develop, as technological innovation increase yields and payouts compensate losses when required. This is however dependent on effective assimilation of financial tools at affordable levels, preventing them from being viewed as higher-interest borrowing mechanisms.

3.2 Demand-Side Basis Risk & Financial Instrument Trust

These social exclusion challenges are further exacerbated by the farmer uncertainty derived from financial basis risk. This can be applicable to stakeholders on the demand and supply-side of these operations, and they are both fundamentally dependent on the effective threshold design in these index insurance tools.

Thresholds that overestimate the required parameter level, for instance excessive precipitation or hurricane windspeeds, can result in scenarios where crops are damaged, but the insurance payout is not triggered. When combined with debt obligations for agricultural technologies, loss of income from lost harvests are felt even more. The Guatemala MAGA Seguro Colectivo Paramétrico Program that monitors and sets thresholds for climate parameters currently operates on insufficient spatial resolutions of 27km by 27km. This is found to potentially result in inconsistent observations and unreliable monitoring in agricultural regions with complex topography, where impacts may be felt significantly but not captured by these tools (World Bank, 2024).

These poor thresholds result in worsened standards of living through lost livelihoods or through collateral damage due to credit defaulting. Furthermore, this defaulting on loans will increase future lender risk and therefore future interest rates, pricing out more of the farmers who need to benefit from these schemes the most. This is likely to occur even in the case of partial collateral reimbursement, which will negatively impact both the currently engaged farmers and discourage future potential stakeholders. Some farmers may have physical inventory or financial reserves to absorb the lost harvest revenue; however, they still will bear negative economic consequences of investing in insurance that doesn't cover their losses, additionally making them less capable of investing in other agri-tech solutions in the future.

In addition to these core demand-side basis risk scenarios, insufficient payout compensation could result in similar worsening of farmers livelihoods. This shows how a misalignment between harvest revenue losses and insurance payouts, even in the case of effective threshold setting can still cause active harm to farmers operations. After implementation, peer effects also become an important factor. Observed negative experiences can spread through tight-knit communities, magnifying lowered adoption rates as long-standing opposition to financial and technological interventions develop.

Therefore, with such downside individual risks, trust in and engagement with these instruments remains low, as farmers opt to continue known and observed traditional practices instead, avoiding unnecessary losses.

3.3 Technology Adoption Reluctance

In addition to financial risk, farmers also experience behavioural barriers with uncertain technologies resulting in a low willingness to adopt unfamiliar practices. The milpa farming practice of intercropping maize, beans and squash is an example of dominant traditional farming practices that have been passed on through generations and engrained in communities' culture (Lopez-Ridaura et al., 2021). With its proven success and limited historical records of anomalous failed harvests, farmers may not fully comprehend the requirement for the effectiveness of other technological innovations that combat climate shocks. Compounding this, they may also be unaware of the increasingly common and extreme nature of these climate events.

This resultant loss aversion is also underpinned by present bias as farmers may discount future financial benefits significantly when compared to the immediate up-front costs, even if they are offset by credit channels, which would instead produce short-term repayment requirements. A relevant example of this is through a misalignment between early-stage costs and longer-term benefits with innovative practices like agroforestry. Tree growing durations and associated output benefits, including better soil structure, lower nutrient stripping and enhanced biodiversity, are often experienced in the longer-term and over multiple harvest periods. This means that present bias may magnify short-term investment costs and hinder technology adoption due to higher perceived risks of these benefits not materialising sufficiently, as there is more time over which these climate shocks can impact farm production. This results in lower perceived aggregate financial benefits for the farmers to capitalise on and with high technological scepticism, adoption also falls.

Similarly, this loss aversion may also be compounded by extreme climate shocks that damage harvest yields regardless of the technology implementation, framing these investments as unnecessary, complex and ultimately financially harmful. This technological risk adds to financial uncertainty with farmers potentially being overloaded with new concepts, products and offerings from external stakeholders who may fail to accommodate to context specific requirements (Mobarak & Rosenzweig, 2013). However, if successful in early years, which, as discussed is most likely to be in the present rather than the future, positive peer effects could accelerate adoption beyond an initial, cautious phase in, as farmers learn from their own and their communities' implementation as seen with more resilient coffee types following the 2013 coffee rust crisis (USDA Foreign Agricultural Service, 2016).

3.4 Informal Structures & Contextual Importance

An important intervening factor for the NGOs bundled finance solutions is the pre-existing context.
One key element here are antecedent informal insurance networks. These are non-binding or contractual in nature but are instead built on social trust and long-term relationships between stakeholders on the same level; that being farmers. When adverse climate impacts reduce harvest yields and revenues, farmers aid their community through transfers of cash, inventory, labour or assets, contributing to individual and community resilience. Whilst these networks may lack

guaranteed assurance, there is evidence that they do reduce demand for formal insurance schemes, including the proposed indexed instruments, as the farmers view these channels are sufficient to cope with risk (Mobarak & Rosenzweig, 2013).

Not only does informal insurance result in reduced demand for its formal counterparts, but they may also bear covariate risk. This is evident when formal insurance mechanisms don't pay out as parametric thresholds are not triggered, but crops are damaged and revenues lost for farmers engaged in both formal and informal schemes.

Alongside this lack of diversification in insurance risk, social dynamics may complicate formal instrument adoption. In the probable scenario that not all farmers switch from informal structures to these formal bundles, for reasons explained in section 3.2, the observation of members opting out of community support schemes could cause social divides and reduce community trust and informal insurance effectiveness. Local insurers and other financial operatives are keen for their existing structures to be integrated into new platforms, rather than a new macro-scale solution replacing them (World Bank, 2024). NGOs will want to work with local stakeholders to avoid potentially disruptive consequences and to better their relationships with local communities which will aid their social licence to operate and ultimately effective bundle and technology adoption. This is especially relevant given they are working with lenders and insurers who are perceived as external stakeholders attempting to penetrate existing markets. Failure to adjust operations and effectively integrate these instruments and practices into existing cultures is a core reason for slow agroforestry adoption in global and Central American coffee plots (Wienhold & Goulao, 2023). Fostering strong relations will help them develop effective operations as exemplified by informal insurance networks, materialising their expected improvements in crop yields, protection and farmer livelihoods.

Finally, any specific acute climate shocks immediately preceding the bundles' introduction will likely have a negative impact on adoption as farmers will be recouping losses from informal networks or will have lower financial reserves, potentially translating to higher perceived risks of technology engagement.

3.5 Closing Demand-Side Remarks

Evidently there are several demand-side factors that limit farmers willingness and ability to integrate into this planned NGO scheme, with them being modulated by climate change through its systemic reach. As climate change increases the probability of extreme events, static index insurance payouts become more likely, assuming effective thresholds are set, but these damaging basis risk scenarios also increase in probability further exacerbating farmer uncertainty. Critically, as outlined in the introduction, thresholds will require regular and consistent updating as climatic event probabilities evolve, but also as community resilience and social development occurs. The extent to which this is done effectively will have systemic impacts on all stakeholders involved, but the magnitude of any negative effects are likely to disproportionately hit farmers as the poorest, most vulnerable involved.

These challenges are also not independent, as they interact and compound for example with farmers that are technological and financially uncertain, already embedded in informal support networks, and therefore may prefer to remain exposed to damaging climate risks without accruing financial liabilities

with little guaranteed positive returns. Experiences like this explain why insurance lacks market penetration accounting for only 1.9% of the national GDP in Guatemala, even before breaking this down into indemnity and index varieties (World Bank, 2024). This will therefore require bolstering by strong public policy support as seen with previous national agricultural policies aimed at increasing commercial development and financial inclusion of smallholder farmers to provide future technical assistance avenues to address these barriers in an integrated way, whilst critically cultivating positive local stakeholder interactions and relationships. (World Bank 2024; Mazviona & Sølvsten, 2025). Evidence from Tanzania (Biffis et al., 2022) and India (Mobarak & Rosenzweig, 2013), further outlined in section 5, show that these financial bundles are achievable and can generate positive impacts, but they require sustained investment in index quality, dedicated community engagement, and product contextualisation.

4 Supply-Side Challenges: Lenders and Insurers

Finance providers face various constraints that could affect their ability to operate locally and finance Guatemalan smallholders' adoption of new agricultural technologies. This section examines how insurers, reinsurers, microfinance institutions, public banks and other stakeholders are affected by various risks including liquidity concerns, basis risk and the local regulatory environment.

4.1 Insurer constraints: correlated risk, data, pricing, and reinsurance

Correlated Risk and Lack of Diversification

Insurers operating in areas exposed to extreme climate events are exposed to correlated risk. While insurance mechanisms traditionally rely on the diversification of risk, climate risks are inherently systemic and correlated. Extreme weather events, such as droughts or flooding, tend to impact entire agricultural areas and can lead to large, aggregated losses. Smallholder farmers in a region will all suffer losses at the same time, so local insurers cannot easily diversify their exposure and may face insolvency risk if a major climate event occurs (World Bank, 2024).

In the Monte Carlo simulation modelled, the higher-quality index produces a higher "default probability" than the lower-quality Scenario 1 index, creating greater losses for the insurer during bad years. Because the farm yields correlate closely with the index, the insurer faces "synchronized losses" instead of many small, uncorrelated claims as the high-quality index triggers massive, simultaneous payouts across the entire region during a drought year. This requires the insurer to either have more capital available or to buy more expensive reinsurance to avoid insolvency. As hazards intensify, insurers may therefore pull back from high-risk markets altogether (World Bank, 2024).

Reinsurance Market

As local insurers in developing markets have limited reserves and capacity to manage highly correlated risk, there has been a need to transfer aggregated risk to international reinsurers. Local insurers will typically front and cede 100% of the risk to the small number of international reinsurers willing to underwrite this class of business (World Bank, 2024). However, In Guatemala and the broader North Central American region, early-stage agricultural insurance programs operate at a small scale and may struggle to generate the premium volumes needed to attract support from international reinsurers. As such, local insurers face increasing pressure to retain more risk, which they are ill-equipped to do. Furthermore, reinsurers rely on catastrophe models that struggle to adapt to rapidly changing climate baselines and correlated extremes. When actual losses diverge from modelled expectations, it generates volatility in pricing and capacity, and reinsurers tend to withdraw from markets when cumulative losses exceed their risk tolerances (World Bank, 2024).

Basis Risk: Technology, Data and Cost Concerns

Insurers require vast amounts of high-quality, long-term meteorological and agronomic data to accurately price premiums and payouts (OECD, 2021). This is particularly relevant for parametric mechanisms such as weather index insurance, where greater access to meteorological data allows insurers to price risk more accurately. This data limitation could therefore lead to basis risk. Furthermore, farmers are reluctant to pay market premiums where they have experienced or observed basis risk occur with peers. In some cases, when only a small number of farmers bought

insurance, financial institutions could not earn enough profit to make continued provision of insurance a worthwhile investment for them (J-PAL 2024).

As shown in the model, If the basis risk causes a false negative, farmers lose trust and disadopt the product, mathematically showing how basis risk destroys the insurer's client base and could prevent the innovative product from reaching scale across the population. An approach to mitigating this risk could be the implementation of on-the-ground audits to verify areas where geo-spatial misalignment occurs. Modelled in Scenario 5, the simulation shows that implementing on-the-ground audits at a cost of $850 would successfully reduce basis-risk dis-adoption by 25% and boosts farmer trust recovery by 50% (Appendix: Figure 4). However, the operational costs insurers would bear to physically verify losses could effectively eliminate the primary cost-saving advantage of parametric index insurance.

4.2 Lender constraints: portfolio risk, enforcement, and incentive compatibility

Portfolio Risk

From a portfolio perspective, these institutions are heavily exposed to climate sensitive activities. Most borrowers are smallholders cultivating maize, beans and coffee in contiguous areas, which restricts diversification by crop and geography (FAO, 2018; Root Capital, 2014). Droughts in Guatemala's Dry Corridor therefore generate correlated income shocks, with many farmers facing losses and liquidity shortfalls simultaneously and lenders experiencing spikes in portfolio at risk (World Bank, 2024). Index and parametric insurance schemes have been introduced, but rising basis risk under climate change means payouts may not trigger when both farmers and lenders incur their largest losses, so insurance cannot be treated as a reliable hedge (Carter et al., 2016). Lenders consequently remain cautious about financing smallholders' investments in new technologies.

Enforcement

Enforcement constraints reinforce this caution. Many smallholders operate under informal or customary land tenure and lack formal property documentation, so collateral is weak (Méthot, 2018; FAO, 2018). Even when assets are pledged, legal proceedings are slow and costly, and large-scale foreclosure after a drought is politically and socially unattractive (Wenner, 2007; World Bank, 2023). Expected recovery rates on defaulted rural loans are therefore low, and lenders respond with high interest margins, short maturities and restrictive eligibility criteria, effectively rationing credit to borrowers and sectors perceived as risky (Wenner, 2007; World Bank, 2023).

Incentive compatibility

Incentive mechanisms are likewise limited. Rural financial inclusion initiatives show that good repayment performance is rewarded with larger loans, slightly lower interest rates or more flexible repayment schedules, and that group lending and cooperative structures are favoured because peer monitoring supports timely repayment (FAO, 2018; NCBA CLUSA, 2024). Some institutions link credit access to participation in technical assistance or financial literacy programmes (NCBA CLUSA, 2024; Center for Financial Inclusion, 2024). However, because insurance cover is viewed as imperfect and collateral remains weak, lenders must hold sizable buffers against potential climate related losses, which limits their capacity to offer substantially better terms to borrowers adopting innovative technologies (Wenner, 2007; World Bank, 2024).

Overall, concentrated climate exposed portfolios, fragile enforcement and constrained incentive instruments create a structural reluctance to extend and adapt credit in support of technology adoption by Guatemalan smallholders. When drought is the principal driver of default and index insurance is perceived as subject to high and rising basis risk, lenders have little assurance that claims will be paid in the seasons when borrowers are most likely to fail, and thus view financing new technologies as adding risk to already vulnerable portfolios (World Bank, 2024; Carter et al., 2016).

4.3 Operational constraints: distribution, servicing, and claims settlement

Distribution channels

Agricultural credit and insurance for smallholders in Guatemala are delivered through public programmes, commercial and cooperative banks, microfinance institutions and non-governmental organisations partnering with private insurers (World Bank, 2023; OECD, 2021). These actors operate thin rural branch and agent networks and rely on occasional outreach via municipal centres and producer associations (OECD, 2021; Center for Financial Inclusion, 2024). High travel costs and security concerns limit direct contact, so enrolment often occurs through batch registration by local leaders, generating errors in client lists and weak alignment between insured units and farm plots. Such distribution constraints create mismatches between the population assumed in index design and those actually insured, increasing perceived basis risk.

Client servicing

Servicing and communication frictions add a further layer. Many smallholders face documentation barriers and low familiarity with formal financial products, while literacy and numeracy can be limited in some indigenous and rural communities (OECD, 2021; Center for Financial Inclusion, 2024). Time-pressed front-line staff frequently lack specialised training on parametric insurance concepts, so farmers' expectations about triggers and payouts often diverge from contractual conditions. For lenders, incomplete or outdated client data and weak after-sales support create uncertainty about who is insured and under which terms. These gaps generate "expectation risk": even where the index tracks area losses, farmers and lenders may still interpret non-payment as product failure.

Claims settlement

Claims settlement practices in Guatemala's parametric schemes reinforce these perceptions. Public and donor-supported programmes use rainfall indices, enrol beneficiaries via lists from municipalities, cooperatives or financial institutions, and pay through public bank branches, automated teller machines or digital codes (World Bank, 2024; WFP, 2021). Evaluations report delays between shocks and trigger communication, mismatches between beneficiary lists and current borrowers, network problems that prevent redemption, and disputes when visible damage occurs but the index does not trigger (World Bank, 2024; J-PAL, 2024). Users rarely distinguish these operational failures from downside basis risk, since the observable outcome is a missing payment in bad years. Coordination problems among insurers, lenders and public agencies, and the lack of integrated systems linking loan balances, coverage and triggers at client level, further hinder the systematic use of payouts for debt service, weakening insurance as a credible credit enhancement tool (World Bank, 2024; WFP, 2021).

4.4 Regulatory Constraints: Strategic role of public policy and safety nets

The supply-side challenges to the innovative product's implementation in Guatemala are further compounded by the local regulatory environment. The Superintendencia de Bancos (SIB), the country's local insurance regulator, has yet to enact specific laws dedicated to parametric insurance and currently takes a discretionary, reactive approach to approving index-based insurance scheme. This is primarily because existing national insurance laws were drafted for traditional indemnity models which require physical damage assessments, while index-based products rely on proxy variables like rainfall thresholds (World Bank, 2024). It's therefore important for future models to consider the impact that formal legislation could have on pricing, scale and predictability of the local insurance and reinsurance market.

Furthermore, the bundled insurance solution may be impacted by the lack of legislation and standardisation around which datasets or other inputs are used for insurance schemes. Farmers' assessment of damages and payouts may conflict with insurer-based estimates, particularly where basis risk materialises. The model illustrates the severity of this risk, noting that a single "False Negative" event, loss without a payout, can trigger a 55% disadoption rate among farmers (Appendix A1: Figure 1). To mitigate this, insurers are forced to apply a farm-scale correlation coefficient discounted by 15 percentage points to account for idiosyncratic microclimate variations not captured by satellite data. Without regulatory supervision and agreement, however, this technical adjustment may cause additional frictions between insurers and uptakers.

Finally, although the *Esfuerzo Seguro* program, which bundles index insurance with loans from BANRURAL, and the *Seguro Colectivo Paramétrico* program, where MAGA acts as the policyholder and pays 100% of the premiums on behalf of the farmers, exist, these solutions are unlikely to achieve the scale needed to cope with needs of all local farmers. (World Bank, 2024). Therefore, the government may need to also incentivize private insurers and lenders to provide the bundled insurance scheme through mechanisms such as tax incentives.

4.5 Implications for bundled product design

The interaction between insurer, lender and operational constraints sets clear limits on what a bundled credit-insurance product can achieve in Guatemala and identifies the frictions that design choices must address. Correlated drought risk, rising basis risk and limited reinsurance capacity confine insurers to simple index products, while concentrated loan portfolios, weak collateral and cautious incentive schemes make lenders reluctant to treat insurance as reliable credit enhancement unless payouts are timely, predictable and tied to debt service. Fragmented distribution, servicing gaps and imperfect claims settlement add further perceived basis risk, as administrative failures often result in missing payments in bad years. These constraints call for examining prior index-insurance experiments to identify which features and strategies have overcome similar barriers and can inform the next section's design choices.

5 Evidence from Similar Experiments: Lessons Learned

Drawing on field experiments across developing countries, this section examines what comparable bundled insurance and credit initiatives have achieved in practice. It distils key lessons around uptake mechanisms, basis risk, trust, and institutional requirements, informing the design recommendations for Guatemala's smallholder context in the subsequent sections.

5.1 Mechanisms for increasing uptake

Three mechanisms shaped by the experiments are particularly relevant: the timing and structure of premium payments, the level at which insurance payouts are directed, and the fragility of adoption in the absence of early positive experience.

5.1.1 Liquidity, timing and the "bundling penalty"

A field experiment in Malawi offered farmers a choice between standard uninsured loans and loans bundled with weather index-based insurance at actuarially fair rates (Giné & Yang, 2009). The authors found that bundling insurance reduced loan uptake by 13% relative to uninsured loans, highlighting a potential "bundling penalty" where farmers interpreted the premium on bundled insurance as a higher effective interest rate (Giné & Yang, 2009). This discouraged uptake for smallholders due to constrained liquidity and cash scarcity at the time of planting crops, presenting a trade-off between upfront payments with input purchases (Giné & Yang, 2009).

An experiment in Ethiopia demonstrated how the "bundling penalty" can be reversed through liquidity-sensitive design through testing the timing of insurance payments with index insurance bundled with credit and input vouchers in the Rift Valley zone (Belissa et al., 2025). Delaying payment was found to increase insurance adoption by roughly 19%, whilst bundling delayed premium insurance with credit and input vouchers increased uptake by 32% and significantly boosted investment in fertiliser and improved seeds (Belissa et al., 2025). Evidence from Kenya and Uzbekistan reinforces this pattern. Bulte et al. (2020) found that subsidised insurance bundled with certified seeds in Kenya increased adoption and crowded-in investments in fertiliser, hired labour and machinery. Moritz et al. (2025) found that in Uzbekistan, insured farmers were 59 percentage points more likely to invest in fertiliser, and that having received a payout in a previous season further boosted fertiliser investment by a further 24 percentage points. Insurance adoption also rose from 56% in the first round to 71-89% in subsequent rounds, suggesting that experiential learning builds trust and sustained participation over time (Moritz et al., 2025).

A key consideration for the NGO is timing of payment. If premiums are charged upfront at the time of planting, the bundle may suppress rather than increase demand for credit. A viable design might require delayed or post-harvest premiums, financing mechanisms or early-stage subsidies to avoid the "bundling penalty". The evidence from Uzbekistan, suggests that experiential learning through repeated rounds of insurance and payout exposure may be a tool for driving sustained participation. However, this highlights the connection to basis risk, that if the product fails to perform credibly in earlier season, liquidity-sensitive design will be ineffective in sustaining uptake.

5.1.2 Unlocking credit supply through meso-level insurance

A common assumption with bundled insurance products, is that the primary barrier to adoption is insufficient farmer demand for insured credit. In Northern Ghana, a randomised field experiment investigated agricultural loans bundled with index insurance, where payouts were directed either to farmers (micro-level) or to lenders (meso-level) to cover loan defaults in the event of drought (Mishra et al., 2021; 2023). The experiment found that meso-level insurance significantly increases loan approval rates by 17-25% driven by reduced lender risk exposure (Mishra et al., 2021; 2023). Crucially, neither micro-level nor meso-level insured loans significantly increase loan application rates, confirming that the inhibiting factor was supply-side risk rationing rather than insufficient farmer demand for credit (Mishra et al., 2023).

These findings suggest that the supply-side risk aversion from lenders is likely to be a barrier to expanding input finance, as well as the need to address concerns and trust on the demand-side to ensure farmers adopt insured credit. A bundled insurance product in Guatemala should therefore incorporate meso-level insurance to protect lenders' portfolios, whilst also ensuring that farmers perceive tangible benefits from the product. Design must account for both demand-side and supply-side, to prevent insurance expanding credit supply without translating to higher uptake and technology adoption for smallholder farmers.

5.1.3 Investment responses and adoption fragility

Beyond initial uptake, a key question is whether insurance promotes sustained investment over time. A multi-year field experiment in Tanzania and Mozambique tested a bundled product that combined drought-tolerant maize seeds with satellite-based index insurance (Boucher et al., 2024). Farmers who experienced a severe shock and received a payout, subsequently expanded their cultivation area by approximately 67% and intensified input use in the following season (Boucher et al., 2024). This demonstrates a "resilience dividend" with insurance compensating for losses whilst also accelerating investment. Farmers who purchased the bundled insurance but did not experience a shock and therefore did not receive payout frequently abandoned the technology in subsequent periods, showing that adoption was highly fragile (Boucher et al., 2024).

The fragility of adoption in Tanzania and Mozambique highlights an area of concern that should be considered for Guatemala (Boucher et al., 2024). During the initial years of having the bundled insurance, farmers who pay for the product but experience no drought and therefore receive no payout may interpret the premium for the bundle as a loss or unnecessary cost, especially if their neighbours who may not have paid the premium are better off. Factors such as risk aversion and present bias may amplify the risk of this in Guatemala for smallholder farmers. The NGO could provide sustained, early-stage subsidies for an experiential phase in the first two to three seasons to gain farmers' trust in the product. Additionally, investment in educating farmers on the benefits of insurance as a protection measure against catastrophic climate events would aid understanding and improve trust of farmers.

5.2 Addressing basis risk and trust

Adoption fragility is ultimately driven by basis risk: when the index fails to capture a farmer's actual losses, trust collapses and participation unravels. The following subsections examine how spatial mismatch, excluded perils, and audit mechanisms have been used to manage this risk across international programmes, with direct implications for Guatemala.

5.2.1 Mitigating Spatial Mismatch and Securing Trust

In Kenya's Kilimo Salama programme, automated solar-powered weather stations were deployed to collect highly localised, objective rainfall data, practically eliminating geographical mismatch and fostering trust among previously sceptical farmers (International Finance Corporation, 2010). Experimental evidence from Kenya quantitatively supports this mechanism by showing that in the case that farmers trust the objective measurement of the insurance product, unconditional input investments increase significantly (Bulte et al., 2020). Specifically, offering free multi-peril insurance bundled with certified seeds increased farmers' total non-seed investments, such as fertiliser and hired labour, by almost 1,700 Kenyan Shillings, representing a statistically significant increase of 0.15 standard deviations (Bulte et al., 2020).

While Kenya successfully utilised highly localised ground-based weather data to foster farmer trust, Guatemala's sparse station network forces domestic programmes to rely on satellite imagery (World Bank Group, 2024). Guatemala's main domestic initiative, MAGA's Seguro Colectivo Paramétrico, currently uses ERA5 satellite data at a coarse 27x27 km resolution, which fails to capture the country's complex topography and microclimates, generating significant spatial basis risk (World Bank Group, 2024). Transitioning to CHIRPS (Climate Hazards Group InfraRed Precipitation with Station data), a high-resolution satellite dataset operating at approximately 5.5x5.5 km that has been successfully applied to characterise drought conditions in the region, would substantially reduce this mismatch (World Bank Group, 2024). However, improved data quality is a necessary precondition for trust rather than sufficient on its own. Since satellite-derived indices are more abstract for farmers, the NGO must pair data quality with transparency of process, through publicly accessible payout registers at cooperative and municipal level, so that farmers can independently verify triggers and payouts, replicating through transparency what Kenya achieved through visible on-site infrastructure (Bulte et al., 2020).

5.2.2 Local Basis Risk and the Limits of Financial Incentives

Beyond spatial mismatch, farmers are acutely sensitive to local basis risk, which occurs when a policy explicitly excludes non-weather perils such as pest infestations (Freudenreich & Mußhoff, 2018). A framed field experiment with Mexican maize farmers revealed that index insurance plagued by local basis risk performed noticeably worse than full indemnity insurance in driving the adoption of hybrid seeds (Freudenreich & Mußhoff, 2018). Furthermore, a 50% premium subsidy completely failed to increase adoption under the index insurance scheme, whereas it succeeded under the indemnity scheme (Freudenreich & Mußhoff, 2018).

The Malawi case also reveals a deeper demand-side issue beyond liquidity constraints. Because highly impoverished farmers already had "implicit insurance" through limited liability, meaning they faced few consequences for defaulting on a standard loan during a drought, forcing them to purchase formal insurance was perceived purely as an unnecessary cost rather than a protective benefit (Giné & Yang,

2009). In Guatemala, where legal enforcement of smallholder collateral is slow and costly (Wenner, 2007; World Bank, 2023), borrowers face similarly weak consequences for default, replicating the implicit insurance dynamic Giné and Yang identified in Malawi. This suggests that even well-designed products can fail if farmers do not perceive tangible value relative to the informal coping mechanisms they already rely upon.

Guatemalan farmers face multifaceted biological threats beyond drought, including coffee leaf rust and pest outbreaks (World Bank Group, 2024). The Malawi case offers a crucial warning: where lenders cannot seize collateral, borrowers already possess implicit "limited liability," making formal coverage feel redundant and suppressing demand (Giné & Yang, 2009). In Guatemala, weak collateral enforcement and insecure land tenure leave most smallholders in the informal economy, only around 8% access formal credit (FAO, 2018; World Bank Group, 2024), presenting a structurally similar dynamic. The NGO should therefore not treat insurance as a mandatory upfront cost. Viable alternatives include subsidised premiums (World Bank Group, 2024), harvest-linked delayed payments (Belissa et al., 2025), or meso-level insurance protecting the lender's portfolio directly without requiring a separate farmer premium (Mishra et al., 2021). To address local basis risk gaps, the bundle should additionally incorporate resilient seed varieties or agronomic training (Bulte et al., 2020; Boucher et al., 2024).

5.2.3 Enhancing Trust Through Audits and Premium Structuring

Managing the perception of basis risk and overcoming severe liquidity constraints are essential for sustaining trust over time. In Ethiopia, a randomised control trial demonstrated that allowing farmers to delay premium payments until after the harvest and bundling index insurance with credit and an input voucher surged baseline uptake by 32 percentage points (Belissa, Lensink & Marr, 2025). To maintain trust and prevent defaults under this delayed premium scheme, the Ethiopian programme instituted an independent agronomic audit to transparently verify losses whenever farmers complained the index had failed them (Belissa, Lensink & Marr, 2025). This transparent process kept defaults remarkably low, with only 29 out of 459 purchasing households, approximately 6%, defaulting (Belissa, Lensink & Marr, 2025). This is supported by findings of a multi-year trial in Tanzania and Mozambique, where farmers who received an insurance payout after a severe yield shock exhibited a "resilience dividend," significantly increasing their input expenditures by $129.50 and expanding their maize area by 1.5 hectares in the following season (Boucher et al., 2024).

To effectively manage basis risk disputes and ensure fairness when the index fails to capture localised shocks, the Guatemalan programme should incorporate conditional, ground-level agronomic audits directly into the contract (World Bank Group, 2024). By combining this transparent fail-safe audit with a delayed premium structure and input vouchers, the NGO can build the necessary demand-side trust to drive high adoption rates and secure long-term agricultural resilience.

To conclude, this evidence does not offer a single replicable model, but it narrows the design space considerably: the products that have succeeded combine transparent, high-quality indices with premium structures that respect farmers' liquidity constraints, institutional mechanisms that rebuild trust when the index falls short, and sustained early-stage support that allows experience to accumulate before commercial viability is expected. These conditions directly inform the model and recommendations developed in the following section 6.

6 Policy Implications and Design Recommendations

To test the design recommendations developed in section 5, an original Monte Carlo simulation parameterised on Guatemala's Dry Corridor context was built. The model [2] evaluates ten design scenarios for the bundled product across six dimensions simultaneously: basis risk (false-negative and false-positive rates), farmer income gain, tail risk (Value-at-Risk and Conditional VaR at 5%), insurer viability (loss ratio), credit enhancement, and drought-tolerant seed adoption over five seasons. Each scenario is run across 5,000 simulated farmers over 200 seasons using a bivariate normal framework to capture the correlation between on-farm rainfall and the index trigger. Key parameters are drawn from CHIRPS calibration data, BANRURAL lending rates, and WFP Dry Corridor cost benchmarks, with the farm-scale correlation coefficient discounted by 15 percentage points following Biffis and Chavez (2017). The model does not claim to forecast outcomes; it provides a scenario-specific, Guatemala-parameterised comparison that the empirical literature alone cannot supply, and it is explicit about its limitations throughout. The outputs are summarised in Figures 1-5 in the Appendix.

6.1 Feasibility assessment

The model gives a clear answer on feasibility: the bundled product is not viable at unsubsidised market rates, but it becomes financially meaningful for farmers under a fifty percent premium subsidy, combined with a CHIRPS-quality index and conditional audits. The next paragraphs set out the conditions that make this possible and the constraints that still bind.

Under fair actuarial pricing with a thirty-five percent commercial loading factor, every scenario without subsidy produces negative income for the farmer. The ERA5 baseline (S1) gives a net loss of nineteen dollars per hectare per year; CHIRPS without subsidy (S2) still leaves a loss of thirteen dollars (see Figure 1, Appendix). This is not a modelling quirk. With a commercially viable loss ratio of 0.74×, the premium of roughly eighty-four dollars per hectare under CHIRPS exceeds the expected payout value after loan interest. That is consistent with J-PAL (2024) finding that uptake at market prices rarely reaches twenty percent, and it confirms that subsidy is a structural prerequisite, not an optional add-on.

A fifty percent premium subsidy under CHIRPS with conditional audits (S10) changes the result: income rises by twenty-nine dollars per hectare per year, while per-season default falls to 1.0 percent and lifetime default over twenty seasons to 20.7 percent. The insurer still stays commercially viable at a loss ratio of 0.74×, and the NGO's annual cost is fifty-five dollars per farmer, made up of a forty-two dollar subsidy and thirteen dollars in audit costs. For forty thousand insured smallholders, that implies about 2.2 million dollars per year, which sits within the range of multilateral agricultural finance programmes. The full scenario comparison across all six output dimensions is provided in Figure 1 (Appendix).

The model points to five enabling conditions. The index must reach at least $\rho \geq 0.65$ at farm scale (Biffis and Chavez, 2017): ERA5 ($\rho_{tam} = 0.30$) falls well short, CHIRPS ($\rho_{tam} = 0.63$) comes close but does not

[2] GitHub repository containing the full implementation of the model, along with instructions to reproduce the results and details on the experimental setup: https://github.com/PauloBrl/Bundling-Crop-Insurance-with-Innovative-Production-Technologies/tree/main

quite meet it, and the NDVI-SPI composite (ρ_{tam} = 0.70) clears it. A formal lender, such as BANRURAL or a comparable rural cooperative, must take part for the Carter et al. (2016) portfolio effect to operate, whereby risk is diversified across a large pool of borrowers, allowing lenders to reduce exposure to idiosyncratic shocks and extend credit more confidently. MAGA extension services must keep distributing drought-tolerant seed for at least three seasons to allow the technology adoption trajectory to develop. Multi-year NGO funding of at least five seasons must be committed in advance, because a programme that withdraws subsidy after one or two seasons will generate a trust collapse rather than a graduation toward self-financing. Finally, the regulatory framework must accommodate interlinked insurance-credit contracts, which Guatemala's current insurance legislation, drafted for traditional indemnity products, does not yet explicitly permit.

Three constraints are binding. Guatemala's station network, with 97 stations over 109,000 km², is too sparse for meso-level calibration below departmental scale. The model's ρ = 0.78 for CHIRPS is also borrowed from Mozambique, because no Guatemala-specific estimate exists; if the true Dry Corridor value is closer to 0.60 or lower after the farm-scale discount, feasibility weakens sharply. Finally, the climate non-stationarity stress test (S9, pd = 50%) produces a lifetime default probability of 100 percent and a VaR of minus eighty-seven dollars per hectare, showing that the current design would fail under a large shift in drought frequency. The income distributions underlying these tail-risk figures, including the bimodal ERA5 distribution produced by its high false-positive rate, are shown in Figure 2 (Appendix). Regular recalibration is therefore essential.

6.2 Product design choices/recommendations

The ten scenarios cover the main design dimensions at once, so the recommendations below rest on quantitative comparison rather than general principle.

Index choice and minimum viable correlation threshold

The key decision is which rainfall index to use. ERA5 at 27x27 km gives a farm-scale ρ of 0.30, far below the Biffis and Chavez (2017) threshold of 0.65, a false-negative rate of 14.5 percent, and a five-percent Value-at-Risk of twelve dollars per hectare. CHIRPS at 5x5 km improves ρ to 0.63, lowers false negatives to 10.4 percent, and raises VaR to thirty-one dollars. The NDVI-SPI composite reaches ρ_{tam} = 0.70, clears the minimum threshold, and gives a false-negative rate of 9.6 percent and VaR of thirty-four dollars. The NGO should treat CHIRPS as the minimum acceptable index and move to the composite within three seasons once NDVI integration is in place. Staying with ERA5 is not conservative; it is a decision to deliver a product that will fail in roughly fourteen out of every one hundred drought years (Figure 1, Appendix, panels 2-3 show the FN and FP rates across all ten scenarios).

Premium timing and harvest-linked deduction structure

The Malawi experiment (Giné and Yang, 2009) showed that collecting the premium at planting, when liquidity is tightest, reduced uptake by thirteen points. Evidence from Ethiopia shows that delayed or harvest-linked collection increased adoption by about nineteen percent and raised fertiliser and seed investment by thirty-two percent. The practical implication is simple: the premium should be deducted from harvest-season loan repayment, not charged at planting. Under S10, the farmer pays forty-two dollars per hectare, the fifty-percent co-payment on the fair premium of eighty-four dollars,

at harvest rather than at loan origination. That avoids the bundling penalty seen in Malawi and matches the payment to the season when the farmer has either received revenue or received a payout.

Meso-level interlinkage versus standalone product structure

Carter, Cheng and Sarris (2016) show that insurance linked to the lender's portfolio creates a 1.39x income multiplier relative to standalone insurance by reducing the risk premium in the lending rate. In the model, that appears as a credit-rate cut from twenty-two to 17.3 percent under CHIRPS (S2, S10), which already includes an eight-dollar per hectare interest saving (Biffis et al., 2022). The mechanism only works if drought payouts are routed at least partly to loan repayment rather than paid entirely as cash. A pure lender-first structure gives no visible benefit to farmers, while a purely cash-based payout weakens the lender's incentive to participate.

Subsidy design: the S7 versus S10 trade-off

This is the most consequential design choice the NGO must make, and the model presents it as a policy judgement rather than a technical optimum. On every quantitative dimension, full subsidy (S7) strictly dominates fifty-percent subsidy (S10): income gain of seventy-one versus twenty-nine dollars per hectare, per-season default of 0.3 versus 1.0 percent, twenty-season lifetime default of 5.3 versus 20.7 percent, and VaR of 116 versus 73 dollars. Year-five adoption is essentially identical at 42.5 versus 42.4 percent, confirming that the adoption ceiling is determined by trust and seed availability, not by the residual premium level within the subsidised range. Figure 4 maps all ten scenarios against NGO annual cost per farmer and farmer income gain, making the S7/S10 trade-off visible at a glance. The quantitative case for S7 is unambiguous. The sustainability case for S10 rests on considerations the model does not capture. A zero co-payment (S7) creates no contractual relationship between farmer and insurer, generates no renewal habit, and makes the programme entirely dependent on uninterrupted NGO funding with no pathway to graduation. The forty-two dollar annual co-payment in S10 establishes a commercial relationship, creates a demonstrated willingness-to-pay signal for future market development, and allows the NGO to begin a subsidy step-down trajectory in Years 3 to 5 without catastrophic demand collapse. The honest position is that S7 is the superior welfare programme and S10 is the superior development finance programme. The NGO must state this trade-off explicitly in its donor reporting.

Basis risk management: conditional audits versus Basis Risk Fund

The model compares S4 (ERA5 plus BRF) and S5 (CHIRPS plus conditional audits). Without subsidy, both produce negative income gains, but the BRF adds two dollars per year to NGO costs without changing adoption, while audits add thirteen dollars per year and lift year-five adoption from 23.1 to 36.4 percent (Figure 4, Appendix, illustrates this clearly: S4 and S5 are nearly indistinguishable on cost but S5 sits materially higher on the adoption axis). The audit premium comes from the trust-recovery effect in Mobarak and Rosenzweig (2013): each objective audit strengthens confidence across the peer network. The BRF, by contrast, pays individuals but does not build community trust and could be depleted quickly in a severe El Niño. The conditional audit protocol is therefore the better basis-risk tool.

Addressing peril basis risk: the named-peril rider

The model's false-negative rate captures only weather-index mismatch, not losses from fall armyworm, coffee leaf rust, local flooding or frost. De Janvry et al. (2014) show that farmers tolerate geographical basis risk more readily than local uninsured losses, and that excluding non-weather perils depresses adoption even when the insurance is otherwise attractive. The NGO should therefore include a named-peril rider for fall armyworm and Roya from Year 1 in pilot areas, with the rider fully subsidised in Years 1 and 2 to remove early friction.

Trust-building mechanisms: the guaranteed payout and peer network

Mobarak and Rosenzweig (2013) show the problem clearly: farmers in strong informal networks see no obvious benefit in non-drought years and may view the premium as a loss. Three steps can break that pattern. First, a guaranteed first-season payout gives the product an immediate visible payoff. Casaburi and Willis (2018) show this works better per dollar than subsidy alone in low-trust settings. Second, a public payout register, at municipal and cooperative offices and online, makes each trigger and payout easy to verify. Third, two or three respected farmer champions per village can spread trust more efficiently than a broader marketing campaign.

Drought-tolerant seed complementarity: why bundling is not optional

Comparing S2, where CHIRPS is bundled with drought-tolerant seed and no subsidy, with S8, where CHIRPS is bundled with conventional seed and no subsidy, can make it look as though drought-tolerant seed lowers welfare. That is the wrong reading. The larger loss in S2 reflects the higher fair premium on a larger insured input package, not a failure of the seed itself (Figure 1, Appendix, panel 1 shows S2 and S8 side by side on the income gain axis). Once subsidy is added, S7 delivers a seventy-one dollar per hectare gain because drought-tolerant varieties raise yields in both severe and normal drought years. Without them, the insurance protects a weaker crop and the resilience dividend described by Biffis et al. (2022) does not emerge. Drought-tolerant seed should therefore be mandatory in the input loan, not an optional extra.

6.3 Implementation Roadmap

The roadmap follows a widely observed development sequence of validation, trust-building, and institutional graduation (e.g. World Bank; CGAP). The phase boundaries are indicative, but the escalation triggers in section 6.5 define when the programme can move on.

Years 1-2 (Pilot and Index Validation)

The first two seasons are for validation and trust-building, not scaling. The NGO should work with 500 to 1,000 farmers in three to four municipios in Chiquimula and Jalapa, where drought exposure is strongest and which overlap with the WFP-Aseguradora Rural pilot footprint, outlined in section 2.2.1. CHIRPS should operate as the main payout trigger, with ERA5 recorded in parallel for comparison. NDVI should be fed in alongside agronomic yield monitoring at twenty sample farms per municipio to build the ground-truth dataset needed for composite calibration. Subsidy should be eighty percent in Years 1 and 2, above the fifty-percent S10 level, so the first cohort is very likely to see a positive outcome. Figure 5 shows why this front-loading matters: adoption growth is slowest in Seasons 1 and 2 regardless of index quality, which is precisely the window in which high subsidy and a guaranteed payout must hold farmers above the fragility threshold. The named-peril rider for fall armyworm and

Roya should be fully subsidised. Baseline data should be collected at enrolment on income, adoption, weather readings and risk preferences.

Years 2-4 (Scaling and Subsidy Step-Down)

If the pilot meets the index-quality threshold (validated ρ ≥ 0.65), delivers at least one visible payout, and retains more than seventy percent of farmers in Year 2, the program can expand to Jutiapa, Zacapa and El Progreso. Subsidy then steps down from eighty percent to fifty percent at the start of Year 3. BANRURAL enters as the lending partner under a tripartite agreement with the insurer and the NGO, activating the Carter et al. (2016) portfolio effect. Conditional audits become formalised, with a maximum thirty-day response time. A second lender, ideally a rural cooperative federation, should join in Year 3 to keep pressure on lending rates. NDVI-SPI can become the main trigger once the pilot calibration dataset is large enough to validate it empirically.

Year 4 and beyond (Institutional Sustainability)

From Year 4 onward, the program should begin a planned step-down toward a long-run subsidy target of twenty-five to thirty percent, financed through government agricultural support, climate-adaptation finance and a levy tied to falling insurer losses as drought-tolerant seed adoption rises. An Index Calibration Committee, INSIVUMEH, MAGA, the insurer's actuary and an independent academic, should review triggers each year. An independent impact evaluation should report in Year 4 so donors can fund later tranches on performance rather than process. The program is not expected to become fully self-financing within five years without public support, but the goal is to cut NGO cost per farmer from fifty-five dollars in Year 3 to below thirty dollars by Year 6. As Figure 5 illustrates, the composite index (S3) produces the steepest adoption curve from Season 2 onward, which is the expected payoff from the NDVI integration work in Years 1-2.

6.4 Conditions for success

The following seven conditions are necessary: if any one of these fails, the program becomes either financially unviable or institutionally weak.

1. Index quality must reach ρ ≥ 0.65 at farm scale. Biffis and Chavez (2017) set this as the minimum for positive welfare effects among risk-averse farmers. ERA5 (ρ_{tam} = 0.30) fails outright and should be replaced by CHIRPS (ρ_{tam} = 0.63) as a transition index and the NDVI-SPI composite (ρ_{tam} = 0.70) as the target standard.
2. Formal lender participation must be real, not assumed. The NGO should check this each year; if lenders keep the benefit as margin instead of passing it through, the twenty-nine dollar per hectare gain under S10 is overstated and the feasibility case weakens.
3. MAGA extension services must remain continuous. Drought-tolerant seed distribution depends on the MAGA network, which has been interrupted before by fiscal tightening. The adoption model assumes uninterrupted supply; if extension stops in any season, adoption will fall below the 42.4 percent Year-5 projection and the actuarial gain will shrink.
4. A multi-year funding commitment of at least five seasons must be secured before launch. Boucher et al. (2024) show that adoption is fragile. If subsidy ends too early, farmers are left with a bad first impression before trust has time to build. Funding should therefore be structured as a multi-tranche facility with performance triggers, not annual renewals.

5. At least one visible trigger payout must occur within the first two seasons. The model's trust-recovery parameter only compounds if farmers see a payout. Mobarak and Rosenzweig (2013) show that non-drought seasons are where disadoption starts. If Seasons 1 and 2 are both dry-free, the NGO should consider a guaranteed first-season transfer as part of the insurance cost.
6. Regulatory approval for the interlinked contract structure must be in place before launch. Guatemala's current insurance law, written for traditional indemnity products, does not clearly allow parametric triggers or routing payouts to loan repayment. The NGO should work with the Superintendencia de Bancos in Year 1 to secure a sandbox-style exemption, as Mexico did for CADENA (Componente de Atención a Desastres Naturales).
7. A climate recalibration protocol must be in place before the first El Niño, not after it. The high-stress scenario (S9, p_e = 50%) shows that a sharp shift in drought frequency drives lifetime default to one hundred percent and VaR to minus eighty-seven dollars. Trigger thresholds based on the current thirty percent drought probability will under-trigger if conditions shift, so the Index Calibration Committee must meet every year and escalate automatically if drought frequency exceeds 40 percent over any three-season window.

6.5 Monitoring, evaluation, and learning

The MEL framework rests on five KPIs, an annual recalibration schedule, an audit trigger protocol and a two-tier default monitoring system that keeps actuarial and programme-assessment metrics separate.

Key Performance Indicators

The five primary indicators and their numerical targets are as follows.

- False-Negative Rate: target ≤ 10% by the end of Year 2 and ≤ 8% by the end of Year 4. CHIRPS is the baseline at 10.4%, and the composite target is 9.6%. Any season above 14.5%, the ERA5 level, should trigger an immediate review (Figure 1, Appendix, panel 2). World Bank DRIFCA (2024) recommends treating basis-risk incidents as an early reputational indicator, not just a financial one.
- Year-5 DT Seed Adoption Rate: target ≥ 35% (the model's central estimate under S10 is 42.4%; the conservative lower bound is about 30%). The full adoption trajectory, including sensitivity to disadoption rate assumptions, is shown in Figure 5. Adoption should be split by gender and indigenous status. WFP Guatemala and IFAD INSURED (2022) report that women make up about eighty percent of insured farmers in current programmes and nearly thirty percent are indigenous, so those groups need close tracking.
- Insurer Loss Ratio: target 0.70-0.80x, with 0.74x as the S10 value. A ratio above 1.0× means the index is over-triggering and becoming unsustainable; below 0.60x suggests under-triggering and reputational risk from uncompensated farmers. Annual loss-ratio review should be the main actuarial control.
- Per-Season Default Probability: target ≤ 1.0% (S10), with 3.0% as the red-flag threshold (the S1 level). This is the actuarial metric used for pricing review and should be tracked quarterly with lender portfolio-at-risk data, then reconciled with the model annually.
- Five-Year Horizon Default Probability: target ≤ 25% over twenty seasons, with S10 at 20.7%. This is the program-assessment metric used for donor reporting. It is calculated as

$P(default\ ever) = 1 - (1 - p_s)^T$ and should be reported each year using observed lender data as the program matures.

Index Recalibration Schedule

The Index Calibration Committee should meet every October after harvest to compare CHIRPS rainfall with agronomic yield data from the twenty sample farms in each municipio. If estimated ρ falls below 0.60 in any department for two straight seasons, the committee must adjust the trigger within thirty days or pause new enrolment there. If drought frequency exceeds 40 percent over any three-season window, p_e must be updated immediately and the fair premium recalculated.

Audit Trigger Protocol

A conditional agronomic audit is triggered when (a) CHIRPS does not breach the trigger in a season and (b) more than fifteen percent of surveyed farmers in a municipio report yield losses above twenty-five percent of expected harvest. If both conditions hold, MAGA sends an independent agronomic team within thirty days to compare ground yields with satellite vegetation data. Findings are reported publicly within forty-five days and posted to the payout register. If the audit shows basis risk affected more than twenty percent of insured farmers, the Index Calibration Committee meets in emergency session and the NGO can use a reserved trust fund, capped at ten percent of annual premium revenue, for a discretionary supplemental payment. The protocol closely follows the Ethiopia programme, where transparency cut the threatened defaulting group to 29 out of 459 farmers (6.3 percent).

Two-Tier Default Probability Monitoring Framework

The model gives two different default metrics, and they should not be mixed up in reporting. Per-season default probability is the actuarial metric for pricing review. Under S10 it is 1.0 percent, and the insurer should compare it with the 3.0 percent adjustment trigger. Lifetime default probability over a twenty-season horizon is the donor-facing metric for household welfare. Under S10 it is 20.7 percent, meaning about one in five enrolled farmers will face at least one negative-income season over a typical five-year programme window. That figure should be reported honestly to donors; it is not failure, but a reminder that the product only partially covers weather risk. The quasi-experimental evaluation design, comparing credit-only, insurance-only and bundled cohorts against the existing MAGA Seguro Colectivo Paramétrico as a natural comparison group, following the Mishra et al. (2023) Ghana protocol, provides the counterfactual needed to assess whether 20.7 percent is an improvement over the uninsured baseline.

Taken together, the five KPIs, annual recalibration, conditional audits and two-tier default monitoring system create a learning loop designed to catch failure early, separate actuarial problems from design problems, and build the evidence base needed for decisions on continuation, subsidy step-down and geographic expansion.

7 Conclusion

Guatemala's smallholder farmers face a compounding vulnerability that no single instrument can resolve: the same climate shocks that destroy harvests also invalidate the informal risk-sharing on which most households depend, while the cost of investing in more resilient technologies remains beyond what constrained credit markets can support. This report has argued that a bundled product combining index-based crop insurance with input credit and drought-tolerant seeds is not merely theoretically attractive, but operationally feasible under defined conditions, and operationally dangerous if those conditions are not met. The demand-side evidence reveals that affordability, trust, and basis risk are not separate barriers but a single compounding dynamic: an index that fails farmers once destroys more value than no insurance at all. The supply-side evidence confirms that lenders will not extend credit on better terms unless insurance credibly shifts their tail risk, which in turn requires the index quality threshold the current ERA5-based programme does not reach. The simulation model quantifies this precisely: subsidy is not an optional enhancement but a structural prerequisite, index quality below $\rho = 0.65$ at farm scale produces a product that harms more than it helps, and the full-subsidy scenario dominates quantitatively while the fifty-percent co-payment scenario dominates on programme sustainability grounds, a tension that must be stated honestly rather than resolved by sleight of hand.

The central recommendation is therefore S10: CHIRPS as a minimum index, a fifty-percent premium co-payment deducted at harvest, conditional agronomic audits as the basis-risk governance mechanism, drought-tolerant seeds bundled as mandatory inputs, and a meso-level interlinkage with BANRURAL that routes payouts partially to loan service. This design delivers a twenty-nine dollar per hectare income gain, holds per-season default at one percent, and produces a commercially viable insurer loss ratio of 0.74x, but only under an eighty percent subsidy in Years 1 and 2, a validated ρ, regulatory approval for the interlinked contract, and a multi-year funding commitment secured before launch, not after the first El Niño. Whether Guatemala's Dry Corridor becomes a replicable model for climate-resilient smallholder finance in Central America, or another pilot that collapsed before trust had time to build, will depend less on product design than on the institutional durability of those preconditions.

8 Appendix A1: Model Outputs

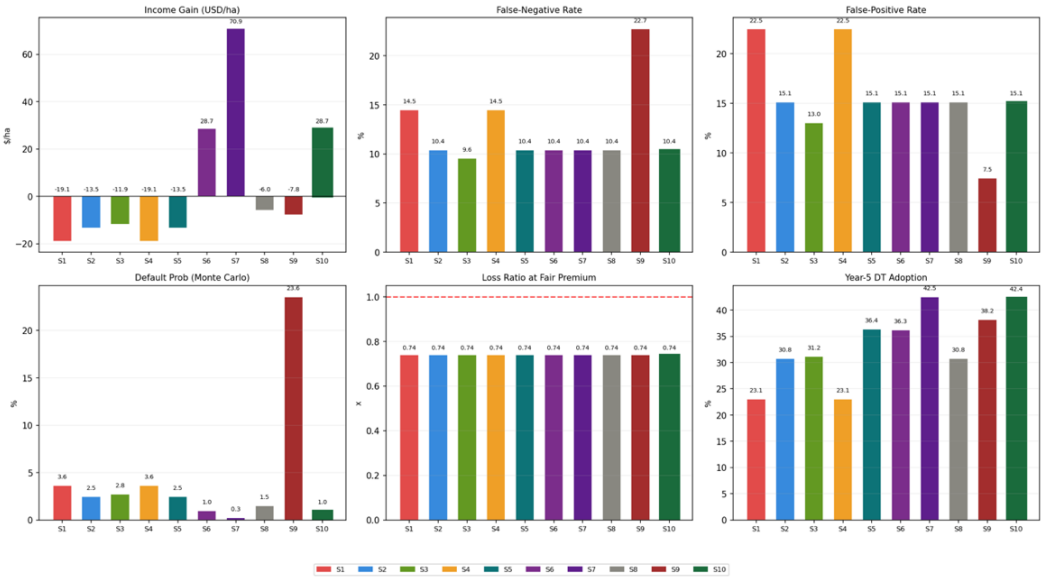


Figure 1: Monte Carlo Evaluation of Bundled Insurance Scenarios: Income, Basis Risk, Default, and Adoption Metrics

This figure constitutes the central output of the model. It presents a side-by-side comparison of all ten scenarios across six key performance metrics, allowing for an immediate assessment of which designs are effective and which are not.

The income gain (top-left panel) indicates that, in the absence of subsidies, the bundled product imposes a net cost on farmers rather than delivering benefits: scenarios S1 through S5 all yield negative gains. Positive outcomes only emerge when a 50% subsidy is introduced (S6 and S10). This result is consistent with the findings of Biffis et al. (2022) in Tanzania, who show that actuarially fair premiums can offset the benefits of insurance unless external support is provided.

The false-negative (FN) and false-positive (FP) rates (top-middle and top-right panels) capture basis risk, which lies at the core of the challenge addressed in the brief. ERA5 exhibits the highest FN rate (14.5%), reflecting its coarse spatial resolution (27 km) and its inability to detect localised drought events. Switching to CHIRPS reduces the FN rate to 10.4%, illustrating the value of improved data quality.

Default probability (bottom-left) represents the key metric from the lender's perspective and directly addresses supply-side concerns. The loss ratio (bottom-middle) demonstrates the financial viability of the insurance product: all scenarios yield a ratio of approximately 0.74x, well below the 1.0x break-even threshold. This corrects earlier model versions (v7), where implausibly high loss ratios (4-9x) indicated a specification error.

Finally, the adoption rate (bottom-right) shows that the most effective design (S10) increases uptake from an initial 23% to 42% over time, suggesting that the bundled solution can achieve meaningful traction among farmers.

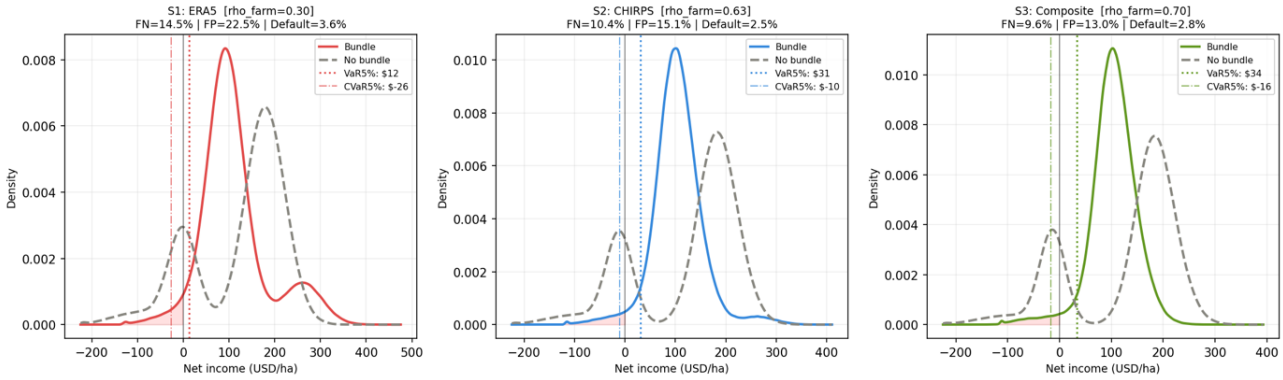


Figure 2: Income Distribution and Downside Risk Under Alternative Index Designs: Monte Carlo Evidence (VaR and CVaR)

This figure presents the full distribution of farmer income under three index designs: ERA5, CHIRPS, and a composite index. The solid lines represent outcomes with the bundled product, while the dashed grey lines correspond to the no-bundle counterfactual. The red shaded area below zero indicates states in which the farmer defaults.

Under ERA5, the income distribution exhibits a bimodal shape, with two distinct peaks. This pattern is driven by its relatively high false-positive rate (22.5%), which generates random payouts in the absence of actual losses. While this may appear to increase upside potential, these gains are effectively insurer-funded transfers and do not reflect genuine improvements in farmer welfare.

By contrast, the CHIRPS and composite index scenarios produce smoother and more concentrated distributions, indicating more consistent and reliable outcomes. The VaR (5%) and CVaR (5%) markers highlight downside risk. CHIRPS, for instance, delivers stronger protection in the lower tail, with VaR and CVaR values of $31 and -$10 respectively, compared to $12 and -$26 under ERA5.

Overall, this figure makes risk more tangible from both perspectives: for farmers, it illustrates the severity of potential downside outcomes, while for insurers, it reflects the dispersion and predictability of payouts. As such, it directly captures the dual demand- and supply-side considerations emphasized in the brief.

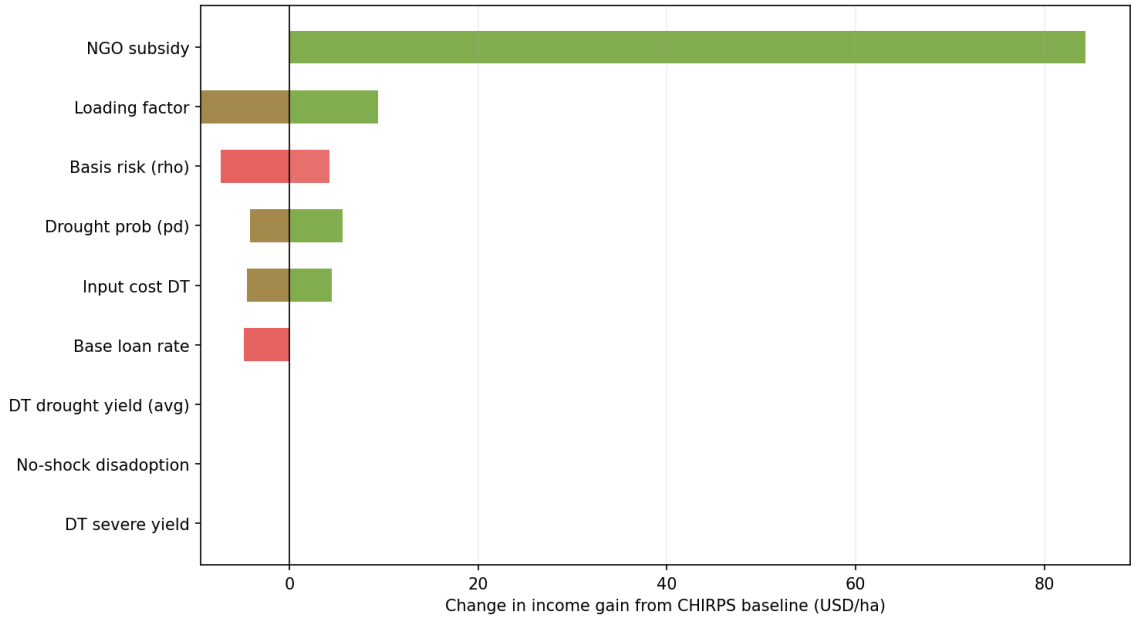


Figure 3: Sensitivity Analysis of Income Gains: Key Drivers of Bundled Product Performance

This figure presents a standard one-at-a-time sensitivity analysis based on the CHIRPS baseline scenario. Each parameter is varied from a low to a high value, and the resulting impact on farmer income gain is measured.

The results clearly indicate that the NGO subsidy is by far the dominant driver, generating a variation of up to $85/ha. This highlights that the most critical lever in the design of the bundled product is the level of premium subsidy, rather than more technical actuarial parameters. The loading factor and the index quality parameter (ρ) emerge as the next most influential variables.

In contrast, agronomic parameters, such as drought-tolerant yield and severe yield outcomes, have only a marginal effect on income gains. This suggests that the viability of the bundled solution is primarily determined by its financial structure rather than by agricultural performance. This constitutes a key insight from a risk management perspective, as it shifts the focus from production improvements to financial design.

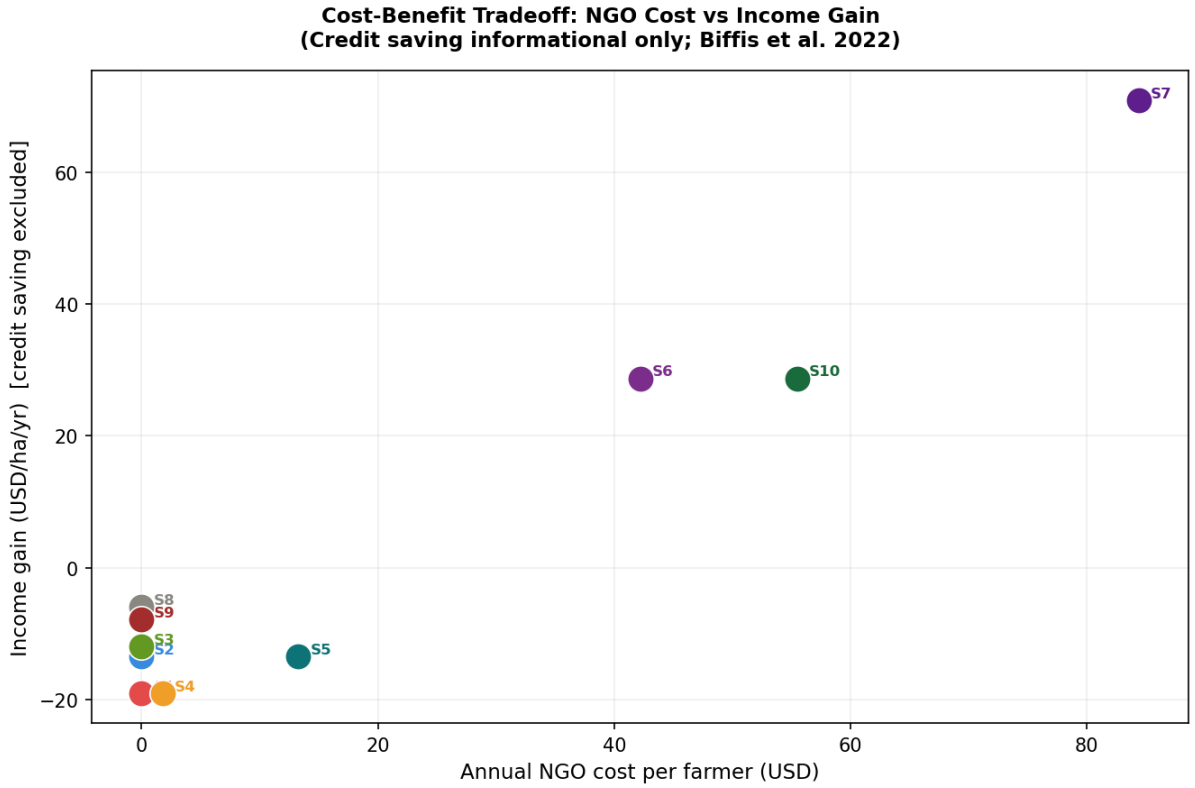


Figure 4: Cost-Benefit Trade off: NGO Cost vs Income Gain (Credit saving informational only; Biffis et al. 2022)

This figure presents a scatter plot of NGO cost (x-axis) against farmer income gain (y-axis), with each point representing a different scenario. The overall pattern is immediately informative. Scenarios S1 to S5 cluster in the bottom-left quadrant, combining low cost with negative income gains, indicating that these designs are ineffective.

By contrast, scenarios S6 and S10 occupy an intermediate “efficient” region, where moderate NGO expenditures generate positive income gains for farmers. Scenario S7 lies in the top-right corner, delivering the highest farmer benefits but at a substantially higher cost, raising concerns about scalability.

A more nuanced comparison between S6 and S10 highlights the value of additional design features. While S10 entails an incremental cost of approximately $13 per farmer per year relative to S6, primarily due to the inclusion of audits, it results in a significantly higher adoption rate (42% versus 36%, as shown in Figure 1). This additional investment appears justified in light of the fragility of early-stage adoption, as emphasized in recent work by Alain de Janvry Boucher et al. (2024).

Overall, this figure functions as a practical decision tool for the NGO, enabling a clear comparison of the trade-offs between financial cost and developmental impact across alternative designs.

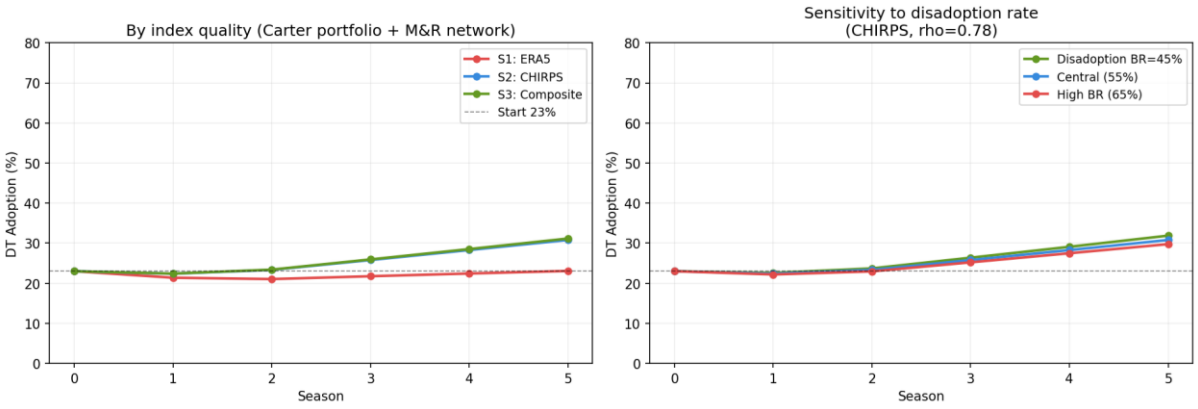


Figure 5: Adoption Trajectory - 5 Seasons. Portfolio channel (Carter 2016) + Subsidy (Biffis 2022) + M&R 2013 network

The left panel illustrates the evolution of technology adoption over five seasons under different index quality scenarios. While CHIRPS and the composite index lead to a gradual increase in adoption, from an initial 23% to approximately 31%, the ERA5 scenario exhibits a near-flat trajectory. This stagnation is driven by its higher false-negative rate, which undermines farmer trust in the insurance product. This mechanism is consistent with the peer effects documented by Ahmed Mushfiq Mobarak and Mark Rosenzweig (2013), whereby farmers who experience unpaid losses transmit negative signals within their communities.

The right panel provides a robustness check with respect to the disadoption rate assumption. Across all specifications, adoption levels remain above the initial baseline, confirming the stability of the results.

Overall, the figure supports the brief's objective of promoting the adoption of climate-resilient technologies. However, it also highlights that adoption gains are gradual and contingent on appropriate product design, particularly in terms of minimizing basis risk and preserving farmer trust.

9 Appendix A2: Model Boundaries and Potential Enhancements

1. Dynamic Premium Repricing

The current model computes the actuarially fair premium once, based on the initial yield-rainfall relationship for drought-tolerant seeds. However, as the share of DT adopters in the insured portfolio grows, the underlying yield distribution shifts insured farmers progressively experience smaller drought-related losses, reducing the insurer's expected payout. A more complete framework would reprice the premium each season as a function of the evolving portfolio composition, generating a feedback loop in which rising adoption lowers the fair premium, which in turn improves affordability and further accelerates adoption. Biffis et al. (2022) highlight this dynamic explicitly, noting that insurers who fail to update actuarial models as seed varieties change will systematically misprice risk, undermining both financial sustainability and farmer trust.

2. Spatial Heterogeneity Across Sub-Regions

The model treats the Dry Corridor as a single homogeneous zone characterised by a uniform drought probability (p_drought = 0.30) and a single index-correlation parameter (rho). In practice, the Corridor spans Pacific lowlands and interior valleys with meaningfully distinct microclimates, and Guatemala's meteorological network comprises only 97 stations across 109,000 km2 (INSIVUMEH; World Bank DRIFCA, 2024). Partitioning the insured population into two or three agro-ecological sub-zones, each with distinct drought frequencies, yield distributions, and rho values, would yield a more realistic assessment of where the product is most and least viable. It would also capture the portfolio diversification benefit available to the insurer when losses in one sub-region do not perfectly correlate with losses in another and would reveal the extent to which an aggregate rho of 0.78 masks substantial spatial variation in index quality.

3. Meso-Level Insurance

The model captures only micro-level insurance, in which payouts flow directly to the farmer. However, evidence from Ghana (Mishra et al., 2021, discussed in Section 5.1.2) indicates that directing payouts to the lender at the portfolio level (meso-level insurance) increased loan approval rates by 17 to 25 percent, by reducing the lender's exposure to correlated default risk. Incorporating a meso-level layer would require modelling the lender's portfolio loss distribution explicitly and demonstrating how meso-insurance compresses the tail of that distribution. This would provide a substantially more convincing representation of the credit enhancement channel than the reduced-form rate formula currently employed.

4. Non-Weather Perils

The model accounts for basis risk arising from spatial mismatch between the weather index and farm-level rainfall (captured through the rho discount) but does not incorporate local perils such as pest infestations or plant disease. Evidence from a framed field experiment in Mexico (discussed in Section 5.2.2) suggests that this form of local basis risk, where losses occur due to perils explicitly excluded from the index contract, constitutes a more significant barrier to trust and technology adoption than geographical basis risk alone. Introducing a stochastic pest or disease shock that causes yield losses independently of the weather index would render the false-negative rate endogenous to the full peril mix, rather than purely a function of weather correlation. This extension would also strengthen the

case for bundling the insurance product with complementary risk-reduction strategies, such as pest-resistant seed varieties or integrated pest management programmes.

5. Heterogeneous Farmers

The Monte Carlo simulation treats all 5,000 farmers as homogeneous agents with identical wealth, risk preferences, and plot characteristics. In practice, smallholder populations exhibit considerable heterogeneity in risk aversion, initial asset levels, and land tenure status. The Mexico field experiment referenced in Section 5.2.2 estimates a mean Constant Relative Risk Aversion (CRRA) coefficient of 1.61 among participating farmers, with significant dispersion. Introducing heterogeneous agents drawn from empirically calibrated distributions of CRRA coefficients and initial wealth would enable the model to identify which farmer segments derive the greatest welfare gain from the bundle and, critically, which segments remain excluded even after premium subsidisation. This distributional analysis is directly policy-relevant, as the NGO must understand whether the product reaches the most vulnerable households or primarily benefits those already near the viability threshold.

6. Bayesian Trust Updating

The adoption component of the model employs fixed disadoption and trust recovery parameters calibrated from the empirical literature. A more theoretically grounded approach would represent farmer beliefs about the insurance product's reliability as a Bayesian posterior, updated each season based on observed outcomes. Under this framework, consecutive correct payouts would shift the posterior upward, while a false-negative event would produce a sharp downward revision whose magnitude depends on the farmer's prior. Such a structure would endogenously generate the adoption fragility documented by Boucher et al. (2024) in Tanzania and Mozambique, without requiring externally imposed disadoption coefficients. Implementation would effectively require a full agent-based architecture, which fell outside the scope of the present analysis but represents a natural direction for future work.

7. Multi-Crop Extension

The report's introduction identifies three distinct farmer segments in Guatemala: maize and bean producers in the Dry Corridor, highland coffee growers, and Pacific coast sugarcane farmers. The model focuses exclusively on the first segment. Extending the framework to coffee would introduce frost risk as a distinct hazard type, longer investment horizons (coffee requires three to four years to reach productive maturity), and exposure to international commodity price volatility. Sugarcane would add hurricane risk with a fundamentally different loss distribution characterised by low frequency but catastrophic severity. A multi-crop model would demonstrate the portfolio diversification benefit available to the insurer across uncorrelated hazard types and would inform the NGO's geographic and sectoral sequencing strategy. This extension constitutes a substantial modelling effort and is best pursued as a dedicated study.

8. Bootstrap Confidence Intervals

The model reports point estimates for tail risk metrics (VaR 5%, CVaR 5%, default probability) derived from the Monte Carlo simulation, supplemented by a seed-stability check demonstrating acceptable cross-seed variation. However, formal confidence intervals are not reported. Implementing a bootstrap resampling procedure over the 5,000 by 200 income matrix, with 90 percent confidence bands around each key performance indicator, would render the statistical precision of the tail

estimates explicit. This is particularly relevant given that the policy recommendation rests on specific point estimates such as a per-season default probability of 1.0 percent, and decision-makers should be informed of the degree of estimation uncertainty surrounding these figures. This extension was not pursued due to scope constraints but would meaningfully enhance the robustness of the reported results.

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